

EDA Summary

Group: 20

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For our project, we will be working with a combined total of 8 datasets. The first dataset is a vehicle listing in the US dataset scraped from Craigslist. 6 out of the 8 datasets are annual registration data of vehicle registrations based on fuel type and zip code in California. The last dataset is a vehicle msrp dataset with vehicles' MSRP values of different brands from multiple tiers. Summary of datasets are as follows:

1. Dataset Summary

vehicles.csv

We will be using **vehicles.csv** for our first dataset and contains 426880 rows and 26 columns of data. However, after filtering listings that are corresponding to CA, we are left with 50614 rows of data and the updated dataframe is saved as **ca_vehicles.csv**. This dataset has vehicle listings from different regions of CA. The features of this dataset we are interested in are 'region', 'price', 'year', 'manufacturer', 'model', 'condition', 'fuel', 'odometer', 'title_status', 'size', 'type', 'state'. From this dataset, 'region' will be mapped to zipcode from vehicle_fuel_type to infer demand in a specific region.

Vehicle_fuel-type_count_by_zip_code_csvs

These registration datasets will be primarily used to create an engineered feature 'demand' using the number of registered vehicles based on zip code across CA. The zip code will be mapped to the regions in the **vehicles.csv**. Through this, we will be able to draw precise vehicle demand across different regions of CA. The features of this dataset we are interested in are 'Zip Code', 'Model Year', 'Fuel', 'Make', 'Duty', 'Vehicles', 'year'. This dataset allows us to understand the trend of vehicles that are registered after six years across various regions in CA. We can also observe the top registered cars and their associated fuel types across different regions over time. All six of the **fuel_type_count_by_zip_code** csvs were merged into one global dataset named **complete_reg.**

car_data.csv

The last dataset is a vehicle MSRP dataset with vehicles' MSRP values of different brands from multiple tiers. The features of interest from this dataset are 'Model', 'Year', 'Status', 'Mileage', 'Price', 'MSRP'. This dataset is essential for calculating and predicting the depreciation that will be used to train the model. This dataset will be mainly used for extracting MSRP values and for estimating MSRPs or most recent prices of similar vehicles.

The target variable for our project : **Price**

Interested features or features to be included in training the ML model:

'region', 'price', 'year', 'manufacturer', 'model', 'condition', 'fuel', 'odometer', 'title_status', 'size', 'type', 'state' from **ca_vehicles**

'Zip Code', 'Model Year', 'Fuel', 'Make', 'Duty', 'Vehicles', 'year' from complete_reg
'Model', 'Year', 'Status', 'Mileage', 'Price' from car_data

2. Preprocessing

vehicles.csv

Starting with the **vehicles.csv**, we removed outliers that are heavily skewing the dataset. In 'price', 'odometer' & 'year', we noticed values that are either unreasonably low and high values. As a solution, we decided to remove listings that are above \$150000 and below \$500 for 'price'. Likewise, we removed odometer readings that are below 1000 and above 300000. Lastly, we only kept the vehicles that were made after 1990. Additionally, we checked for and removed duplicate listings from the dataset to ensure each vehicle entry is unique and does not bias the analysis

complete_reg

For this dataset, we will be label encoding the categorical values in this dataset for heat map visualizations. We then removed 'OTHER/UNK' entries as well. This data does not contain any duplicates.

car_data.csv

For this dataset, we will be removing the first four digits that correspond to year in the 'Model' column of this dataset since this dataset already contains a column 'Year'. Since the 'Price' column and 'MSRP' columns have the similar values, we will be dropping the 'MSRP' column since it has some description of the MSRP value as well. For the 'Price' column, we will be dropping the commas included in the price and consequently cast the column as a float so it's easier to work with. Moreover, we will be cleaning the 'Mileage' column by formatting non numerical values to NaN.

3. Important Measures

complete_reg:		
	Vehicles	year
count	2.129321e+06	2.129321e+06
mean	6.777663e+01	2.021256e+03
std	1.884081e+02	1.611327e+00
min	1.100000e+01	2.019000e+03
25%	1.600000e+01	2.020000e+03
50%	2.900000e+01	2.021000e+03
75%	6.200000e+01	2.022000e+03
max	3.740700e+04	2.024000e+03

Figure 1

vehicle_msrp:			
	Price	Year	Mileage
count	108432.000000	108432.000000	61199.000000
mean	52095.876992	2020.181736	41962.091374
std	29923.533801	3.960148	31832.911103
min	15000.000000	1949.000000	0.000000
25%	31988.000000	2019.000000	18005.000000
50%	44747.000000	2022.000000	35563.000000
75%	61999.000000	2023.000000	59996.000000
max	200000.000000	2023.000000	974302.000000

Figure 2

Figure 1 shows a summary of the number of vehicles registered along with the corresponding year. Figure 2 gives us summary statistics on Price for newer cars along with their odometer readings. A correlation analysis has been performed on the **ca_vehicle** to study the relationship between final listing price, the year and the odometer reading of the vehicle. The correlation analysis reveals that vehicle price has a moderate positive relationship with year (0.53), indicating that newer cars tend to list for higher prices, and a moderate negative relationship with odometer (-0.55), confirming that higher mileage vehicles are associated with lower prices. These findings validate year and odometer as the two primary drivers of used car depreciation in the dataset.

4. Plots & Visualization

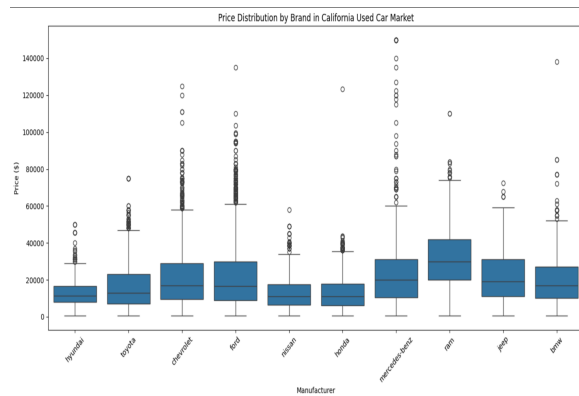


Figure 1

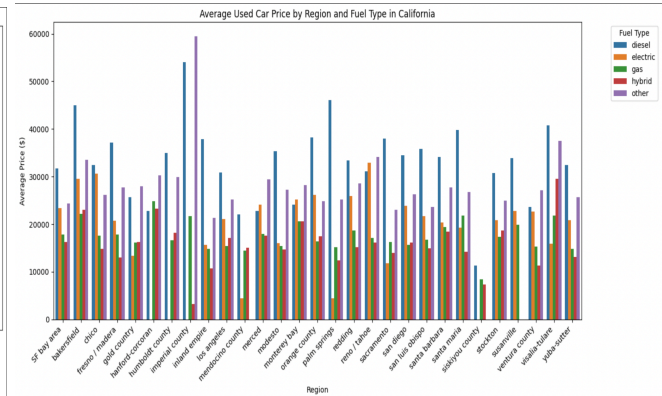


Figure 2

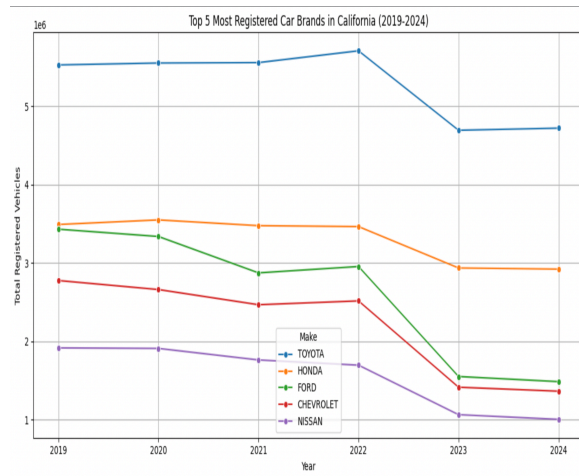


Figure 3

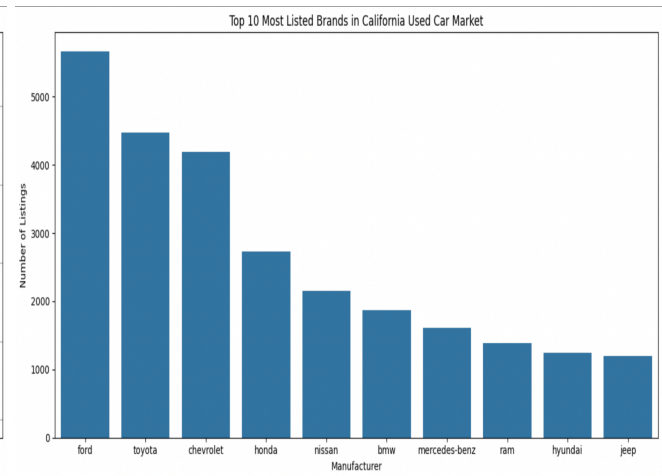


Figure 4

Figure 3 represents the top 5 most registered vehicles in CA. Those top 5 makes appear again as the top 5 most listed used cars, only with different ordering. This suggests that vehicles of those makes will have an easier turnaround time and stronger resale demand. This hints us that those vehicles will depreciate better compared to other brands. According to Figure 1, the resale values of these brands cluster around the median price of \$15,500, indicating they occupy the most active segment of the used car market in California.

5. Preliminary Insights Or Questions

In Figure 2, we uncovered that diesel vehicles are among the highest priced across every region in California. This is an important finding because it reveals that fuel type alone is not a clean signal for depreciation without controlling for vehicle type, as diesel listings are predominantly trucks and heavy duty vehicles which command significantly higher prices than passenger cars. This finding reinforces our decision to incorporate vehicle type and tiering into our modeling approach. Furthermore, the variation in electric vehicle prices across regions, where SF Bay Area electric vehicles list is higher than those in inland regions like Fresno, suggests that geographic demand plays a meaningful role in resale value. We hypothesize that geographic demand, engineered from regional registration counts, will be a significant predictor of vehicle depreciation rates, with areas

of high EV adoption accelerating the depreciation of gasoline vehicles. We also hypothesize that controlling for vehicle type alongside fuel type will produce more accurate depreciation estimates than using either feature in isolation.